Senior Data Engineer

Harshik Joginipelly
Email: harshikjoginipelly@gmail.com LinkedIn: http://www.linkedin.com/in/harshikj
Phone: +1 (773) – 809 – 3040

Professional Summary

- Around 10 years of extensive experience in Data Engineering with strong expertise in data ingestion, modelling, querying, processing, analysis and implementation of enterprise level systems.
- Experience working on development and design of various scalable systems using Hadoop technologies in various environments using Hive, Spark, HDFS, Map Reduce, YARN, Kafka, Pig, Hive, Sqoop, HBase, Cloudera Manager and HortonWorks.
- Experience in analyzing data using Python, R, SQL, Microsoft Excel, Hive, PySpark, and Spark SQL for Data Mining, Data Cleansing and Machine Learning.
- Hands-on experience in the entire project life cycle including data infrastructure building, cleansing, manipulation, engineering, modelling, optimization (performance tuning), QA and data pipeline (ETL) to Data warehouse Deployment On-premise and on Cloud (Azure, AWS).
- Partnered with cross functional teams across the organization to gather requirements, architect, and develop proof of concept for the enterprise Data Lake environments like MapR, Cloudera, Hortonworks and AWS.
- Experience working with systems processing event-based logs, user logs, machines, Azure AD, Azure Data Factory, event hub and event Queue etc.
- Designed and implemented the Medallion Architecture framework, using Azure tools to create structured layers (Bronze, Silver, Gold) for efficient data ingestion, processing, and consumption.
- Hands-on experience building PySpark, Spark Java and Scala applications for batch and stream processing involving Transformations, Actions, Spark SQL queries on RDD's, Data frames and Datasets.
- Experience in data engineering and building ETL pipelines on batch and streaming data using PySpark, Spark SQL.
- Experience in building and optimizing Big Data pipelines, architectures and data sets on building Data Lakes.
- Experienced in building highly scalable Big-data solutions using NoSQL column-oriented databases like Cassandra, MongoDB and HBase by integrating them with Hadoop Cluster.
- Extensive experience working on ETL processes consisting of data transformation, data sourcing, mapping, conversion and loading data from heterogeneous systems like Flat Files, Excel, Oracle, Teradata, MSSQL Server.
- Proficient at writing MapReduce jobs and UDF's to gather, analyze, transform, and deliver the data as per business requirements and optimizing the existing algorithms for best results.
- Sound experience on Normalization and De-normalization techniques on OLAP and OLTP systems.
- Experienced with version control systems like Git, GitHub, CVS, and SVN to keep the versions and configurations of the code organized. Experienced in using Kafka as a distributed publisher-subscriber messaging system.
- Experience with Jira, Confluence and Rally for project management and Oozie, Airflow Scheduling Tools.
- Experience in building interactive dashboards, performing ad-hoc analysis, generating reports and visualizations using Tableau and PowerBI.
- Experienced in writing SQL Queries, Stored procedures, functions, packages, tables, views, triggers using relational databases like Oracle, DB2, and MySQL, Sybase, PostgreSQL and MS SQL server.
- Good experience working in Agile/Scrum methodologies, communication with scrum calls for project analysis and development aspects.
- Experience in possessing an extensive mathematical, analytical skill with strong attention in detailed analysis, makes significant contributions both individually and working as a team.

Technical Skills:

Programming Languages	Python, Scala, SQL, Java, C/C++, Shell Scripting
Python Libraries	Pandas, Scikit-Learn, Tensorflow, Pytorch, Matplotlib
Hadoop Technologies	Hive, Spark, HDFS, Map Reduce, YARN, Kafka, Pig, Hive, Sqoop, HBase, Oozie, Zookeeper, Cloudera Manager and HortonWorks
Cloud Platforms	Azure Cloud, AWS Cloud architecture, AWS Big Data computing

Machine Learning	Decision Tree, SVM, Naïve Bayes, Regression, Classification, Neutral networks
Databases	Oracle, DB2, MySQL, Sybase, PostgreSQL. MS SQL Server
NoSQL Databases	MongoDB, Cassandra, HBase
Visualization Tools	Tableau, Power BI, Advanced Excel
Version Control Systems	Git, GitHub, CVS, SVN
IDEs	PyCharm, IntelliJ IDEA, Jupyter Notebooks, Eclipse

Professional Experience:

Wells Fargo | Minneapolis, MN **Jan 2023 - Current**
Senior Data Engineer

Responsibilities:

- Designed, developed and delivered the Code to IT/UAT/PROD and validate and manage data Pipelines from multiple applications with fast-paced Agile Development methodology using Sprints with JIRA Management Tool.
- Implemented streaming data pipelines using Azure Event Hubs and Databricks to ingest data into Delta tables.
- Built real-time data integration pipelines using Apache Kafka ensuring seamless data flow between disparate systems and real-time analytics capabilities.
- Implemented and managed distributed data processing systems using Apache Spark and Hadoop, effectively handling and storing petabyte-sized datasets.
- Led multiple projects where I leveraged Azure Synapse, resulting in significant improvements in query performance and enhanced analytics capabilities
- Set up comprehensive monitoring and alerting for Kafka clusters using Kafka ensuring timely detection and resolution of issues, which improved system uptime by 20%.
- Extracted, Transformed and Loaded data from Sources Systems to Azure Data Storage services using a combination of Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics.
- Successfully implemented end-to-end Medallion Architecture on Azure, creating structured data layers.
- Automated the ingestion of diverse data sources using Azure Data Factory, ensuring seamless and reliable data movement into the Bronze layer.
- Utilized Azure Databricks for real-time data transformation and processing, cleaning raw data in the Bronze layer and transforming it into structured formats in the Silver layer.
- Performed analysis on existing data flows and created high level/low level technical design documents for business stakeholders that confirm technical design aligns with business requirements.
- Established and managed CI/CD pipelines with Azure DevOps to automate deploying data engineering solutions.
- Worked on building large-scale enterprise data warehousing projects, actively involved in design, architecture and development of ETL pipelines to load data into Azure Synapse using Databricks and Azure Data Factory.
- Implemented lake house architecture powered by Databricks and used features such as Unity Catalog, DLT, Delta Tables, DB-SQL, DBR workflows.
- Created and maintained normalized and denormalized models to facilitate efficient reporting and analysis across various business domains.
- Applied industry-standard best practices for dimensional modeling, ensuring the accuracy and efficiency of data structures.
- Designed and implemented cloud-native data models using Azure Synapse, Databricks, and ADF to optimize analytical workloads.
- Validated and optimized the ETL processes to align with the data model, ensuring the correct and efficient flow of data into data warehouses and Azure Synapse.
- Worked on consuming data from Flat files, JSON, API and stored the data on ADLS.
- Worked on improving performance of Spark applications to bring down cost and SLA.
- Development of machine learning predictive ETL pipelines, leveraging geological and seismic data to optimize exploration processes.
- Achieved 40% improvement in predictive accuracy, enhancing decision-making capabilities for exploration teams. Led a team of six developers to migrate applications.

- Ingested API data into Azure Data Storage services (Blob) using Eventhub and PySpark in Databricks. Data was further transformed into a structured format and stored in DataLake gen2.
- Worked on PySpark and Spark SQL transformation in Azure Databricks to perform complex transformations for business rule implementation.
- Developed and maintained complex PL/SQL scripts, stored procedures, and functions to support data transformation and business logic.
- Integrated PL/SQL-based workflows with cloud technologies like Azure and Snowflake to enhance data pipeline efficiency.
- Leveraged PL/SQL for advanced data manipulation and reporting tasks, contributing to high-quality, actionable insights.
- Designed and optimized Snowflake schema models (Star and Snowflake schemas) to improve query performance and reduce redundant data storage across fact and dimension tables.
- Experienced with SnowSQL for loading data into Snowflake, leveraging its capabilities for efficient data ingestion from various sources, including flat files, JSON, and Parquet formats.
- Implemented advanced query optimization techniques in Snowflake, including materialized views, clustering keys, and micro-partition pruning to improve analytical query performance.
- Developed stored procedures and UDFs (User Defined Functions) in Snowflake using Python and JavaScript to automate data transformation logic and minimize computational overhead.
- Leveraged Snowflake's Time Travel and Zero-Copy Cloning to enhance data recovery, backup strategies, and environment management without additional storage costs.
- Integrated Snowflake with Azure Data Factory and Databricks to create highly efficient ELT workflows, reducing latency in data ingestion and transformation.
- Designed Snowflake-based data pipelines that handle semi-structured data formats (JSON, Avro, Parquet), leveraging VARIANT columns and lateral flattening for optimal storage and querying.
- Maximized query optimization within Snowflake for an external reporting process by bringing down monthly costs by 52% while increasing query speed by 88%
- Implemented RBAC (Role-Based Access Control) and data masking in Snowflake, ensuring sensitive data security and compliance with industry regulations like GDPR and HIPAA.
- Responsible for deployments to DEV, QA, PRE-PROD (CERT) and PROD using Azure.
- Integrated Azure Data Lake Gen2 seamlessly with Azure Synapse, enabling direct querying and analysis of data stored in the data lake using Azure Synapse SQL.
- Effectively managed Snowflake Data Warehouses, with a total capacity of over 2.2PB, ensuring high performance and scalability.
- Migrated data pipelines, from Azure to Snowflake multi-cluster data warehouse as it handles semi-structured data which are in turn connected to analytical engines for reporting.
- Configured and implemented the Azure Data Factory Triggers and scheduled the Pipelines. Created and configured a new event hub with the provided event hubs namespace.
- Leveraged GitHub for version control, collaboration, and deployment automation using GitHub Actions, ensuring seamless integration with Azure environments.
- Leveraged Microsoft SQL Server as a globally distributed, multi-model NoSQL database service for building highly responsive and scalable applications with low-latency data access.
- Leveraged Azure Purview and other data governance tools to ensure end-to-end data governance and management across various cloud platforms.
- Collaborated with cross-functional teams to develop a unified data governance model that aligns with business requirements and complies with internal and external policies.
- Proficient in developing and maintaining data pipelines using Microsoft Fabric, ensuring seamless data flow and processing.

Environment: Python, Pandas, Scikit-Learn, Tensorflow, Pytorch, Matplotlib, Azure Data Lake, Azure Storage, Azure Data Factory, Azure SQL, Azure DW, Azure DevOps, Databricks, Snowflake Schema, Machine Learning, ETL, Hadoop, Spark, Apache Kafka, REST API, PySpark, SQL, Microsoft SQL Server, Power BI, Apache Airflow, Jenkins, Power BI.

Liberty Mutual Insurance | Boston, MA **Sept 2020 – Dec 2022**

Senior Data Engineer

Responsibilities:

- Participated in requirement grooming meetings which involves understanding functional requirements from a business perspective and providing estimates to convert those requirements into software solutions.
- Developed a complex 12-tier hierarchy data model consisting of 1500+ tables as a single source of truth for analysts and ML engineers.
- Integrated Delta tables with other Azure services such as Azure Data Factory for orchestrating data workflows and Azure Synapse Analytics for advanced analytics.
- Implemented Azure Stream Analytics to process streaming data in real-time, ensuring timely ingestion and transformation of data into the Medallion Architecture layers.
- Ensured data security and compliance by implementing Azure Active Directory for access management, Azure Key Vault for secret management, and encrypting data at rest and in transit
- Brought down Snowflake costs by 15% by eliminating runaway costs from poor usage by implementing Snowflake timeouts, warehouse sizing, analyzed accounts, and having monthly meetings with users.
- Acted as a liaison between technical and non-technical teams, translating business requirements into actionable technical tasks and vice versa.
- Created shell scripts to monitor data pipelines, automate data backups, and schedule recurring processes in cloud-based data engineering workflows.
- Integrated bash/ksh scripts with cloud services like Azure Data Factory and Databricks to automate data movement and transformation across platforms.
- Enhanced system automation through shell scripting, reducing the time spent on manual administrative tasks by 30% and improving overall system reliability.
- Utilized SQL Loader for efficiently loading large volumes of data into Oracle databases, ensuring smooth data migration and minimizing downtime during batch processing.
- Integrated SQL loader utilities within cloud environments (e.g., Azure Synapse, Snowflake) to facilitate seamless data migration and loading between on-premises systems and the cloud.
- Led the adoption of Azure Managed Services Stack for data management, leveraging services such as Azure Data Factory, Azure Databricks, and Azure Synapse Analytics to modernize the data architecture.
- Implemented dimensional modeling techniques using Apache Spark to design and optimize data warehouse schemas for analytical reporting purposes.
- Extensive experience in designing data ingestion processes into Azure Synapse using Apache Beam and Airflow.
- Provided mentorship and guidance to junior team members, fostering a collaborative and knowledge-sharing environment.
- Utilized Azure Portal, Azure CLI, Azure Monitor, and Microsoft SQL Server SDKs/tools for monitoring database performance, tracking usage metrics, diagnosing issues, and managing Cosmos DB resources effectively.
- Utilized Terraform for IaC within Snowflake while building out CI/CD pipelines with GitHub Actions for automated deployment and custom code checking.
- Optimized data processing workflows using PL/SQL to enhance performance and reduce execution times.
- Automated data integration tasks by creating reusable PL/SQL code, ensuring efficient and consistent data processing.
- Designed and managed Kafka consumer groups for various applications ensuring balanced load distribution and efficient message processing, resulting in a 40% increase in processing efficiency.
- Proficient in deploying machine learning models (ML)on Azure Functions enabling scalable and reliable inference.
- Performed end-to-end Architecture, implementation assessment of various Azure services like ADF, AF etc
- Worked on Azure Data Pipeline to configure data loads from Azure data factory into Datalake Gen2 and have.
- Developed and deployed Azure Functions services for ETL migration services by generating a serverless data pipeline which can be written to Data factory.
- Extracted data from Data Lakes, EDW to relational databases for analyzing and getting more meaningful insights using SQL Queries and PySpark.
- Implemented AB test computation engine using Python, Snowflake, and parallel processing techniques to deliver precise metric calculations for more than 100 tests each day with an increased computation speed by 40%.

- Worked on Spark improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frame, and Pair RDD's.
- Worked on monitoring and troubleshooting the Apache Kafka-storm-HDFS data pipeline for real time data ingestion in Data Lake in Azure SQL.
- Solved performance issues in Hive and Pig scripts with understanding of Joins, Group and Aggregation and how it translates to MapReduce jobs.
- Managed GitHub repositories for version control, enabling collaborative development and efficient CI/CD pipeline integration with Azure DevOps.
- Worked on PySpark APIs for data transformations created dashboards for Data Visualization using Power BI.

Environment: Python, Pandas, Scikit-Learn, Tensorflow, Pytorch, Matplotlib, Azure Data Lake, Azure Storage, Azure Data Factory, Azure SQL, Azure DW, Azure DevOps, Databricks, Snowflake Schema, Machine Learning, ETL, Hadoop, Spark, PySpark, Kafka, REST API, PySpark, SQL, Power BI, Apache Airflow, Jenkins, Power BI

WellCare | Tampa, FL **April 2018 to Aug 2020**
Data Engineer

Responsibilities:

- Built high-quality Data warehouses and data lakes at enterprise level, worked with cross functional teams to automate data ingestion and schedule jobs at daily, weekly & monthly frequencies on AWS cloud.
- Designed and developed scalable data lake architectures using AWS S3, Glue, and Redshift, enabling efficient data storage, retrieval, and analytics for enterprise-wide use cases.
- Modernized the data warehouse environment by using AWS cloud-based Hadoop system, data manipulations, and maintained tables for reporting, data science, and dash boarding and ad-hoc analyses.
- Utilized Python, Spark on AWS ElasticSearch and Sagemaker to develop & execute Analytics & Machine learning models for fraud detection and risk management.
- Developed ETL Processes in AWS Glue to migrate data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift and Aurora PostgreSQL. Used Python for pattern matching in build logs to format warnings and errors.
- Implemented CDC (Change Data Capture) strategies in AWS Glue and Redshift to ensure incremental data ingestion, reducing data processing overhead and optimizing storage costs.
- Built logical and physical data models for Snowflake for different types of workloads, developed large scale data intelligence solutions around data warehouses.
- Ingested data by applying cleansing and transformations, leveraging AWS Lambda and AWS Glue Step Functions to orchestrate complex workflows.
- Developed and optimized partitioning strategies for AWS Glue and Redshift, leveraging Z-order clustering, distribution keys, and sort keys to improve query performance.
- Built reusable, parameterized ETL pipelines using Glue, Lambda, and Step Functions, enabling flexible ingestion of structured and semi-structured data (JSON, Avro, Parquet, ORC).
- Involved in designing and integrating various system components such as big-data event processing frameworks (Spark), distributed messaging systems(Kafka), and SQL databases like Aurora PostgreSQL for high-performance transactional workloads.
- Implemented data quality, integrity and reliability and creating spark performance tuning solutions through the data pipeline by designing, maintaining, and promoting data governance, enrichment & database efficiency.
- Built the analytics data warehouse and cubes on cloud- based Hadoop, Git version control system, automatic deployment using Jenkins & scheduling systems for job automation.
- Designed and implemented AWS EC2 instances for running scalable computing environments, managing the infrastructure, and handling real-time workloads.
- Built scalable ELT pipelines using dbt (Data Build Tool) on AWS to transform and model data in Redshift, ensuring data quality and reusability.
- Automated data pipeline deployment using Terraform and AWS CloudFormation, reducing manual intervention and improving CI/CD efficiency.
- Utilized AWS ECS to orchestrate and deploy Docker containers for containerized applications, ensuring smooth scaling and management of services.

- Integrated AWS API Gateway with various back-end services, enabling secure and scalable API access to application functionality.
- Leveraged AWS Lambda to run serverless computing functions for processing real-time data streams and event-driven architectures.
- Optimized Aurora PostgreSQL queries and indexing strategies to enhance performance and support large-scale analytical queries.
- Designed high-availability, multi-region replication strategies for Aurora PostgreSQL, ensuring fault tolerance and minimal downtime.
- Developed robust data lineage and governance frameworks using AWS Glue Data Catalog and AWS Lake Formation, ensuring compliance with regulatory requirements.
- Created several complex Reports (sub reports, graphical, multiple groupings, drilldowns, parameter driven, formulas, summarized and conditional formatting) in Power BI for visualization of data.
- Involved in building unit tests and integration tests for automated data validation quality checks before deployment.
- Monitored and logged application performance using AWS CloudWatch, ensuring real-time visibility into system performance and resource utilization.

Environment: Python, AWS (S3, Elastic Search, SageMaker, Glue, Redshift, Aurora PostgreSQL, EC2, ECS, SQS, SNS, Lambda, Step Functions, API Gateway, Event Bridge, CloudWatch, IAM), Machine Learning, Snowflake, Hadoop, Spark, Spark SQL, Scala, HBase, Hive, Pig, Sqoop, MapReduce, HDFS, Kafka, MongoDB, Git, Jenkins, Power BI.

Couth Infotech Pvt. Ltd | Hyderabad, India **July 2014 to Dec 2017**
Data Engineer

Responsibilities:

- Worked with Hadoop Ecosystems components like HBase, Sqoop, Zookeeper, Oozie, Hive and Pig with Cloudera Hadoop distribution.
- Involved in the start to end process of Hadoop jobs that used various technologies such as Sqoop, Pig, Hive, MapReduce, Spark, and Shells scripts.
- Managed and supported enterprise Data Warehouse operation, Big Data application development using Cloudera and Hortonworks HDP.
- Wrote and automated shell scripts (bash, ksh) to handle ETL processes, automate data ingestion, and streamline data transformation tasks in a Linux environment.
- Developed and optimized shell scripts for log processing, ensuring efficient parsing and filtering of system logs for monitoring and troubleshooting purposes.
- Utilized bash/ksh scripting for system automation, scheduling jobs, and orchestrating tasks across different environments, improving operational efficiency and reducing manual intervention.
- Coded ELT data pipelines using Matillion to load data into Snowflake from a variety of data sources including databases, flat files, and API's while transforming datasets to derive business value.
- Improved the performance and optimization of the existing algorithms in Hadoop using SparkContext, Spark-SQL, Data Frame, Pair RDD's, Spark YARN.
- Developed Spark code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- Developed a Spark job in Java which indexes data into ElasticSearch from external Hive tables which are in HDFS.
- Explored with Spark improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frame, Pair RDD's, Spark YARN.
- Imported the data from different sources like HDFS/HBase into Spark RDD and developed a data pipeline using Kafka and Storm to store data into HDFS.
- Used Spark streaming to receive real time data from the Kafka and store the stream data to HDFS using Scala and NoSQL databases such as HBase and Cassandra.
- Utilized PySpark to distribute data processing on large streaming datasets to improve ingestion and process.
- Executed Hive queries on Parquet tables stored in Hive to perform data analysis to meet the business requirements.
- Configured Oozie workflow to run multiple Hive and Pig jobs which run independently with time and data

- Imported and exported the analyzed data to the relational databases using Sqoop for visualization and to generate reports for the BI team. Created several types of data visualizations using Python and Tableau.

Environment: Cloudera, Hortonworks, Hadoop, Sqoop, Pig, Hive, MapReduce, Spark, PySpark, Shell Scripting, SQL, Hortonworks, Python, Java, Scala, HDFS, YARN, Kafka, Zookeeper, Oozie, Cassandra, Tableau, Agile.

Education:

- *Jawaharlal Nehru Technological University, Hyderabad, India* May 2014
  Bachelor of Technology, Information Technology